Digital Traceability for Surgical Systems

Solving the "Dark Data" problem in Eye Surgery Machine Production

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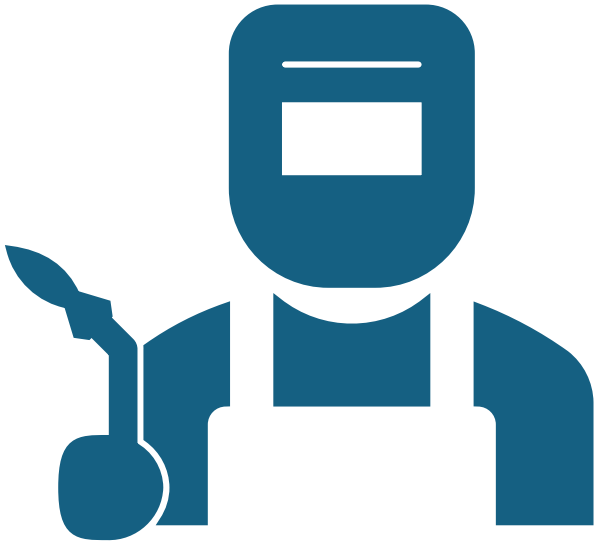
The Current Process (Generation)



- **Step 1:** Assemblers build high-precision modules.
- **Step 2:** Technicians test for pressure, voltage, and accuracy.
- **The Gap: Erp-lex system:** Only records "PASS/FAIL".
  - **Manual Logs:** Actual technical scores are written on **paper**.
- **Concept:** This is **"Dark Data"** – valuable but invisible to the system.

The Business Problem (Efficiency)

- **Assembly Line:** 7 stations (Machine is powered OFF).
- **The Nightmare:** At **Final Test**, if the machine fails, we don't know why.
- **The Bottleneck:** Technicians must search through **100+ pages of paper** to find one score.
- **Result:** High Rework Costs & wasted hours (High MTTR).

The Data Engineering Solution

Lifecycle Alignment:

Goal: Moving from "Searching Paper" to "Data-Driven Decisions".

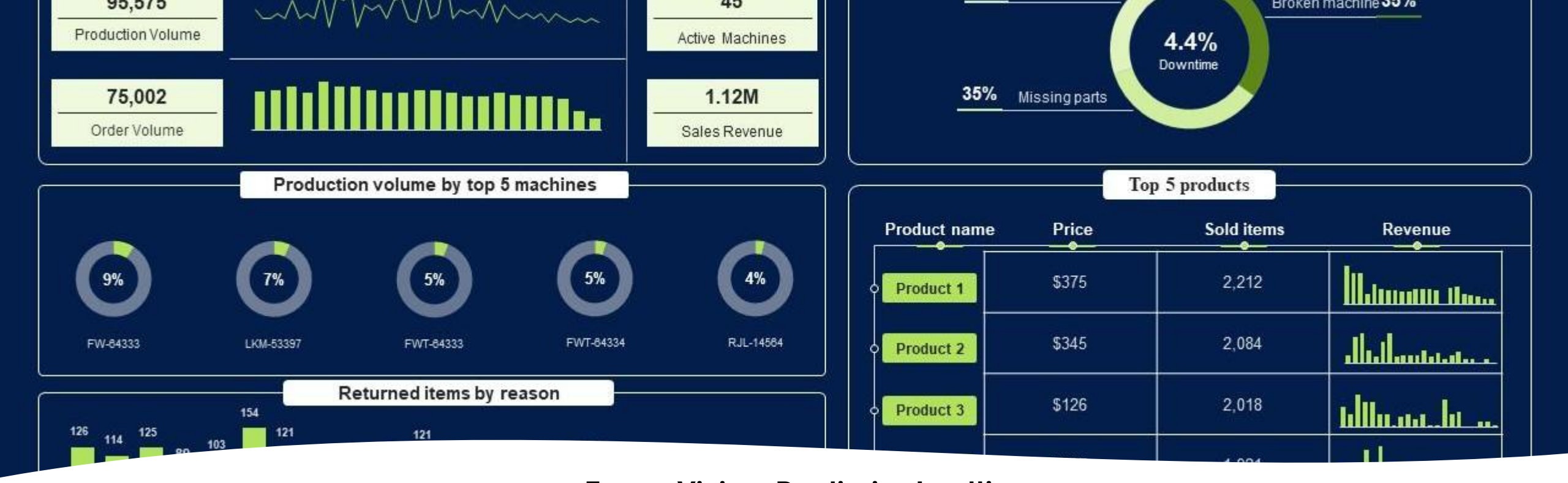
Ingestion: Digitizing paper logs + Erp-lex data.

Integration: Joining data via **Serial Number** to create a **Unified Model**.

Serving: A digital dashboard for instant troubleshooting.

Hidden Challenge: Test Fixture Blindness

- **The Issue:** Technicians rotate shifts; they don't know the history of the testing equipment (Fixtures).
- **The Problem:** A broken fixture fails 4 parts today, and 4 tomorrow. No one notices because data isn't shared.
- **The Impact:** We blame the parts (Modules) when the problem is actually the **Test Fixture**.
- **Solution:** Centralized history reveals fixture failures immediately.



- **Future Vision: Predictive Intelligence**
- **Infrastructure Monitoring:** Detecting when a fixture needs calibration *before* it fails.
- **Cumulative Effects:** Finding "dangerous combinations".
  - *Example:* 3 parts pass individually (low limit), but their **combination** causes a system failure.
- **Pattern Recognition:** Only a digital system can see these hidden risks.

Conclusion & Teamwork

- **Team Roles:** Ingestion (Data Capture)
  - Unified Model (Logic & Joins)
  - Serving (Dashboard & UI)
- **Final Quote:** "Turning Silent Data into a tool for better decisions."
- **Impact:** Faster production, lower costs, and safer machines for patients.

Thank you,

Sayed Amini, 2/27/2026

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